Aarush Ram Anandh

aarush.anandh_ug2024@ashoka.edu.in | +91 9659786783
github.com/MushuRam01 | linkedin.com/in/aarush-ram-anandh

EXECUTIVE SUMMARY

Machine learning researcher with diffusion-model, image-processing and health-data experience.
Contributed to asthma exacerbation detection achieving approximately 70% sensitivity in study evaluation.
Python, C++ and PyTorch applied to image analytics and noisy longitudinal physiological signals.

EDUCATION

Ashoka University — B.Sc. (Hons.), Computer Science and Mathematics **Expected 2028**

RELEVANT RESEARCH EXPERIENCE

Constrained Image-Synthesis Lab, Ashoka University **Sep 2025–Present**
Research Fellow

- Improved latent-diffusion colour fidelity to $\Delta E_{00} < 1$, with approximately **4× better colour accuracy** than evaluated open-source baselines; results reported in the preprint below.
- Developing **Controlled Generalised Sliced Wasserstein (CGSW)**, an optimal-transport framework with control-conditioned probes, projections and slice aggregation; paper submitted to **JMLR**.
- Conducting a **CGSW experiment** on hyperspectral waste images with **224 spectral bands and seven categories**, constructing object-level spectral measures to study class separability.

Koita Centre for Digital Health, Ashoka University **May 2026–Present**
Research Intern (part-time)

- Contributed to a wearable-data pipeline achieving approximately **70% sensitivity (recall)** for **asthma exacerbation detection** in study evaluation.
- Developed preprocessing for **191 patients and 15,691 patient-days**, combining HR/HRV, sleep, temperature and activity data with smoothing, residual and change-point features.
- Built XGBoost/CatBoost and RNN/LSTM modelling workflows; implemented **patient-wise evaluation across five seeds**, separating detection from onset prediction.

COMPUTER VISION EXPERIENCE

Avartan Labs **May–Jul 2026**
Product Engineering / Machine Learning Intern

- Raised rare-class correct detection from **40% to 80%** (**+40 percentage points**) in an **RT-DETR** detection, classification and tracking pipeline for **18 waste-material classes**.
- Investigated focal loss, logit adjustment, class-specific thresholds and targeted fine-tuning for rare classes; evaluated detector failure modes through crop-based refinement.
- Helped set up a **near-infrared waste-segregation system** for visually similar plastics, including parts procurement and preliminary experiments.

SELECTED PROJECTS

CDR-Decomposition Pipeline **Jan 2026**
Vector Graphics / Training Data [GitHub]

- Built a **Python** pipeline to isolate structural die-line vectors from CorelDRAW packaging artwork, supporting generation of **20,000+ training inputs** for lab use.
- Automated headless **LibreOffice** CDR-to-SVG conversion, SVG/XML parsing, raster separation, luminance conversion and geometric filtering, retaining inspection outputs and metadata.

PUBLICATION

Inference-Time Loss-Guided Colour Preservation in Diffusion Sampling (2026).
Public preprint: arXiv:2601.17259; colour-fidelity research described above.

TECHNICAL SKILLS

Programming: Python, C++, Bash. **Frameworks:** PyTorch, Diffusers, TensorFlow, OpenCV.
Methods: Diffusion models, image processing, detection, classification, optimal transport, time-series analysis.